OPTICS · SEMESTER 5 · MODULE 1

Answer key

Formal solutions to all 42 questions of the model paper, written the way an examiner wants to read them.

BEFORE YOU READ AN ANSWER

Each answer starts with a **Know first** line, for example T7 P4 M1 . These are topics in File 1 (Prepare). If any of them feels new, go back and read it first. Then try the question yourself, and only then read the solution.

Mark notes (in grey) show roughly how marks are usually split. They are a guide, not an official scheme.

Section A · Short answers

2 marks each. Write 2–3 sentences. Add a formula or a small figure if it helps.

1 PYQ N20 Q1 · adapted

Define optical path. How is it related to the time taken by light?

KNOW FIRST

T1 Rays and optical path P1 Light, rays and refractive index

The optical path between two points is the product of the refractive index of the medium and the geometrical path length: $L = n\,l$. For a path through several media, or a medium of varying index, $L = \int n\,ds$.

Since light travels at c/n in the medium, the time taken is $\tau = \int \frac{n\,ds}{c} = \frac{L}{c}$. So the optical path is the distance light would travel in vacuum in the same time.

ANSWER $L = \int n\,ds = c\,\tau$

2 PYQ N22 Q11

Explain Fermat's principle of extremum path.

KNOW FIRST

T2 Fermat's principle: stationary, not "least" T1 Rays and optical path M3 Slope zero: finding a stationary point

Fermat's principle: the actual path of a ray between two points is the one for which the optical path $\int n\,ds$ is an **extremum** (or, more generally, stationary) compared with all neighbouring paths: $\delta \int_A^B n\,ds = 0$.

The extremum may be a minimum (straight-line propagation, reflection at a plane mirror, refraction), a maximum (reflection at a concave mirror curving more sharply than the ellipse touching it at the point of reflection), or constant (elliptical reflector). So "least time" is only a special case.

3 PYQ N20 Q11

Write down the three results obtained from Fermat's principle.

KNOW FIRST

T2 Fermat's principle: stationary, not "least" T3 Result 1 and 2: straight lines and the law of reflection T4 Result 3: Snell's law from Fermat

1. **Rectilinear propagation:** in a homogeneous medium light travels in straight lines, since $\int n\,ds = n\times$ length is least for a straight line.
2. **Laws of reflection:** the incident ray, the reflected ray and the normal are coplanar, and the angle of incidence equals the angle of reflection.
3. **Laws of refraction:** the incident ray, the refracted ray and the normal are coplanar, and $n_1 \sin\theta_1 = n_2 \sin\theta_2$ (Snell's law).

4 PYQ N19 Q11

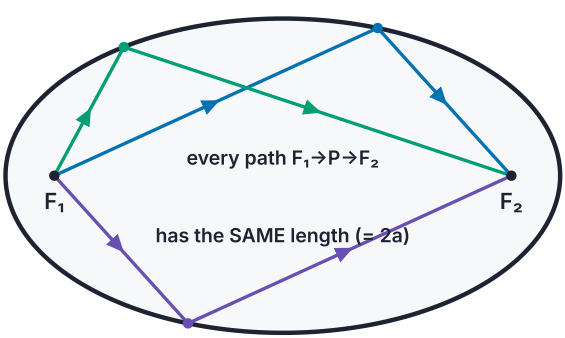
State Fermat's principle of stationary time. Explain with respect to an elliptical reflector.

KNOW FIRST

T2 Fermat's principle: stationary, not "least" T5 Special mirrors and perfect images M5 Ellipse and parabola: the focus property

Statement: a ray of light travels between two points along the path for which the time taken (optical path) is stationary with respect to small variations of the path.

Elliptical reflector: for a point source at one focus S_1 of an elliptical (ellipsoidal) mirror, every point P on the mirror satisfies $S_1P + PS_2 =$ constant. So all paths from S_1 to S_2 via the mirror take equal time. The time is neither a maximum nor a minimum but constant, i.e. stationary. Hence every ray from S_1 reaches S_2 after reflection.



Elliptical reflector. Light from one focus, after reflecting anywhere, reaches the other focus. All the paths are equal, so the path is stationary (neither longer nor shorter than its neighbours).

5

What is meant by a ray of light? When is the ray description valid?

KNOW FIRST

T1 Rays and optical path

A ray is the path along which light energy travels, drawn as a line (normal to the wavefronts). Rays are an idealisation: if we try to isolate a single ray with a very small aperture, diffraction spreads the light.

The ray (geometrical-optics) description is valid when all apertures and obstacles are very large compared with the wavelength, so that diffraction is negligible. Geometrical optics is the limit $\lambda \to 0$.

6

Why does a spherical mirror of large aperture not form a sharp image? Name the defect.

KNOW FIRST

T5 Special mirrors and perfect images T6 Paraxial rays, axis and signs

A spherical surface makes the optical paths from an object point to the image point equal only for paraxial rays (only up to terms in θ^2). Rays striking the mirror far from the axis cross the axis at different points (nearer the mirror for a concave mirror), so a point object gives a blurred patch. This defect is called **spherical aberration**. A paraboloidal mirror is free from it for parallel light.

7

Show by Fermat's principle that a paraboloidal mirror focuses parallel rays to one point.

KNOW FIRST

T5 Special mirrors and perfect images M5 Ellipse and parabola: the focus property T2 Fermat's principle: stationary, not "least"

For a parabola, every point Q is equidistant from the focus S and the directrix: $QS = QL$. Consider rays parallel to the axis, starting from a plane P perpendicular to the axis. Path $= PQ + QS = PQ + QL$, which is the same for every ray (the distance from plane P to the directrix). All parallel rays have equal optical paths to S. So by Fermat's principle they all meet at S.

8 PYQ N22 Q1 · adapted

What is the paraxial approximation? What does it allow?

KNOW FIRST

T6 Paraxial rays, axis and signs M1 Radians and the small-angle trick

In the paraxial approximation we consider only rays that pass **close to the axis** and make small angles with it. This allows $\sin\phi \approx \tan\phi \approx \phi$ and $\cos\phi \approx 1$ (or $1 - \phi^2/2$). We can also measure distances from the pole instead of from the foot of the perpendicular from the point of incidence. Under it, all rays from an object point meet at a single image point.

9

State the Cartesian sign convention.

KNOW FIRST

P4 The sign rule (Cartesian convention) T6 Paraxial rays, axis and signs

With light incident from left to right:

1. All distances are measured from the pole (or the optical centre).
2. Distances measured in the direction of the incident light (to the right) are positive; those against it are negative.
3. Heights above the axis are positive, below are negative.
4. The radius of curvature is positive if the centre of curvature lies to the right of the surface, negative if to the left.

10

Meridional and skew rays.

KNOW FIRST

T6 Paraxial rays, axis and signs

A **meridional ray** lies in a plane containing the optical axis. A **skew ray** does not lie in any such plane. For a single spherical surface the axis is the line joining the object point O and the centre of curvature C. Any ray from O, together with this line, defines a plane through the axis, so every such ray is meridional and, after refraction, crosses the axis again.

11

Air to glass ($n = 1.5$), incidence 30°. Angle of refraction?

KNOW FIRST

T4 Result 3: Snell's law from Fermat P2 Laws of reflection and refraction

1. $n_1 \sin\theta_1 = n_2 \sin\theta_2$
   Snell's law.

2. $\sin\theta_2 = \dfrac{1 \times \sin 30^\circ}{1.5} = \dfrac{0.5}{1.5} = 0.333$
   Substitute.

3. $\theta_2 = \sin^{-1}(0.333) = 19.5^\circ$
   Smaller than 30°: the ray bends towards the normal on entering the denser medium.

ANSWER $\theta_2 \approx 19.5^\circ$

12

What is a virtual object?

KNOW FIRST

T9 Foci of one surface, and surfaces in a row P4 The sign rule (Cartesian convention)

When the rays falling on a surface are converging towards a point that lies **beyond** the surface, that point acts as a virtual object: the rays never actually start from it. In a system of two surfaces, if the image formed by the first surface lies beyond the second surface, the rays reach the second surface before they meet. That image is then a virtual object for the second surface, with $u > 0$ in the Cartesian convention.

13

Focal lengths of a single refracting surface; show $f_2/f_1 = -n_2/n_1$.

KNOW FIRST

T9 Foci of one surface, and surfaces in a row T7 Refraction at one spherical surface (geometry route)

From $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$:

1. First focus: $v \to \infty$, $u = f_1$: $f_1 = -\frac{n_1 R}{n_2 - n_1}$

   Object position whose image is at infinity.

2. Second focus: $u \to -\infty$, $v = f_2$: $f_2 = \frac{n_2 R}{n_2 - n_1}$

   Image of an object at infinity.

3. Dividing: $\frac{f_2}{f_1} = -\frac{n_2}{n_1}$

14

Mirror equation from the refraction formula with $n_2 = -n_1$.

KNOW FIRST

T10 Reflection at a spherical mirror T7 Refraction at one spherical surface (geometry route)

Reflection can be treated as refraction into a medium of index $n_2 = -n_1$, because then Snell's law $n_1 \sin\phi_1 = -n_1 \sin\phi_2$ gives $\phi_2 = -\phi_1$: the ray returns with an equal angle. Substituting in $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$:

$$-\frac{n_1}{v} - \frac{n_1}{u} = -\frac{2n_1}{R} \quad \Rightarrow \quad \frac{1}{u} + \frac{1}{v} = \frac{2}{R}$$

which is the mirror equation.

15 PYQ N24 Q1

Define a thin lens.

KNOW FIRST

T11 Thin lens formula and lens-maker's formula

A lens is a portion of a transparent medium bounded by two spherical (or one spherical and one plane) refracting surfaces. It is called a **thin lens** when its thickness is negligible compared with the object distance, the image distance and the radii of curvature of its surfaces. All distances can then be measured from the centre of the lens.

16

Signs of R_1 and R_2.

KNOW FIRST

P4 The sign rule (Cartesian convention) T11 Thin lens formula and lens-maker's formula

Biconvex	$R_1 > 0$, $R_2 < 0$
Biconcave	$R_1 < 0$, $R_2 > 0$
Meniscus	R_1, R_2 of the same sign (both positive or both negative)

Reason: R is positive when the centre of curvature of that surface lies to the right of it.

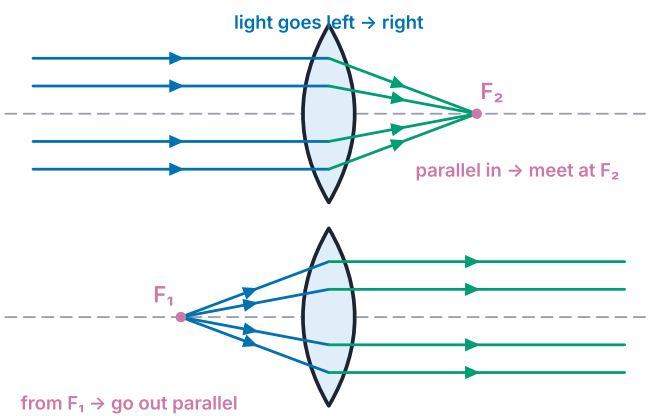
17 PYQ N24 Q2

Define first principal focus of a lens.

KNOW FIRST

T12 Principal foci and focal length T13 Finding the image by drawing

The first principal focus F_1 is the point on the axis such that rays diverging from it (for a convex lens) or converging towards it (for a concave lens) emerge from the lens **parallel to the axis**. Its image is at infinity. Its distance from the lens is the first focal length f_1 (equal to $-f$ for a thin lens in air).



Top: **second principal focus F_2**: rays parallel to the axis meet there after the lens. Bottom: **first principal focus F_1**: rays starting there come out parallel. For a thin lens in air both are a distance f from the lens.

18 PYQ N25 Q1

Glass ($n = 1.5$) biconvex lens in water ($n = 1.33$): converging?

KNOW FIRST

T12 Principal foci and focal length T11 Thin lens formula and lens-maker's formula P1 Light, rays and refractive index

Yes, it still converges. By the lens-maker's formula $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$, where $n = n_{glass}/n_{water} = 1.5/1.33 = 1.128$.

Since $n - 1 = 0.128 > 0$, and $\left(\frac{1}{R_1} - \frac{1}{R_2}\right) > 0$ for a biconvex lens, f is positive: converging. But it is much weaker: $\frac{f_{water}}{f_{air}} = \frac{0.5}{0.128} \approx 3.9$, so the focal length is about four times longer.

ANSWER **Still converging; focal length ≈ 3.9 times larger**

19

Three rules of ray construction.

KNOW FIRST

T13 Finding the image by drawing T12 Principal foci and focal length

1. A ray parallel to the axis passes, after refraction, through the second focus F_2 (or appears to come from it, for a concave lens).
2. A ray through the optical centre goes on undeviated.
3. A ray through the first focus F_1 (or directed towards it) emerges parallel to the axis.

The point where any two of these meet after the lens is the image of the point they started from.

20 PYQ N24 Q3

Define lateral magnification.

KNOW FIRST

T15 Lateral magnification M2 Geometry tools

Lateral (transverse) magnification is the ratio of the height of the image to the height of the object, both measured perpendicular to the axis with their signs: $m = h'/h$. For a thin lens $m = v/u$. A positive m means an erect image and a negative m an inverted one. $|m| > 1$ means the image is enlarged.

21

Newton's formula and its signs.

KNOW FIRST

T14 Newton's formula P4 The sign rule (Cartesian convention)

$x_1 x_2 = f_1 f_2 = -f^2$ for a thin lens in air. Here x_1 is the distance of the object measured from the first focus F_1, and x_2 is the distance of the image measured from the second focus F_2, both with Cartesian signs. Since the product is negative, x_1 and x_2 have opposite signs: an object to the left of F_1 gives an image to the right of F_2. (In magnitudes: $|x_1||x_2| = f^2$.)

22

Show $m = -v/u$ for a spherical mirror.

KNOW FIRST

T15 Lateral magnification T10 Reflection at a spherical mirror

For a refracting surface $m = \dfrac{n_1 v}{n_2 u}$. A mirror behaves like a surface with $n_2 = -n_1$ (Q14). Substituting:

$$m = \frac{n_1 v}{-n_1 u} = -\frac{v}{u}$$

23

Longitudinal magnification $= m^2$.

KNOW FIRST

T15 Lateral magnification M3 Slope zero: finding a stationary point

1. $\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$
   Thin lens formula (f fixed).

2. $-\dfrac{dv}{v^2} + \dfrac{du}{u^2} = 0$
   Differentiate. $d(1/x) = -dx/x^2$.

3. $\dfrac{dv}{du} = \dfrac{v^2}{u^2} = m^2$
   Since $m = v/u$ for a thin lens.

dv/du is always positive, so the image moves in the same direction as the object.

24

Power of a lens; concave lens, $f = 25$ cm.

KNOW FIRST

T12 Principal foci and focal length P3 Mirrors and lenses from Plus Two

The power of a lens is the reciprocal of its focal length in metres: $P = 1/f$. Its unit is the dioptre (D). It measures how strongly the lens converges ($P > 0$) or diverges ($P < 0$).

Concave lens: $f = -25$ cm $= -0.25$ m, so $P = \frac{1}{-0.25} = -4$ D.

ANSWER $P = -4$ **dioptre**

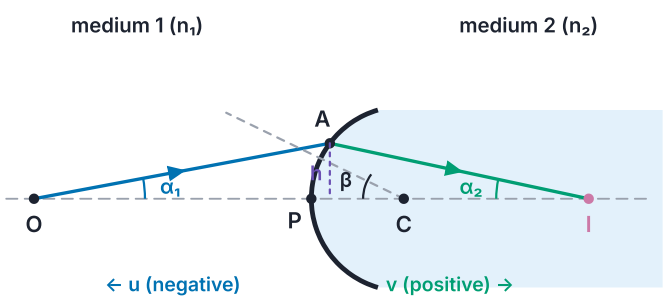
Section B · Paragraph / problem
5 marks each. Figure + steps + result. Say "paradial" wherever you use small angles.

25 PYQ N19 Q18

Derive the condition for image formation by a spherical refracting surface.

KNOW FIRST

T7 Refraction at one spherical surface (geometry route) T6 Paraxial rays, axis and signs P4 The sign rule (Cartesian convention) M1 Radians and the small-angle trick M2 Geometry tools



Refraction at one curved surface. O = object, P = pole, C = centre of curvature, I = image, A = where the ray hits, at height h. Grey dashed = normal (always through C).

A spherical surface of radius r (centre C) separates media of indices n_1 and n_2. A point object O on the axis is at distance x from the pole P. A ray OA meets the surface at A (height h) and, after refraction, cuts the axis at I, at distance y from P. CA is the normal at A. Let ϕ_1, ϕ_2 be the

angles of incidence and refraction, and α_1, β, α_2 the angles that OA, CA and AI make with the axis.

1. $\phi_1 = \alpha_1 + \beta, \quad \phi_2 = \beta - \alpha_2$
   Exterior angles of triangles OAC and ACI.

2. Paraxial rays: $\alpha_1 \approx \dfrac{h}{x}$, $\beta \approx \dfrac{h}{r}$, $\alpha_2 \approx \dfrac{h}{y}$, and $\sin\phi \approx \phi$
   Small angles; the foot of the perpendicular from A is taken at P.

3. Snell's law $n_1\phi_1 = n_2\phi_2$: $\quad n_1\left(\dfrac{h}{x} + \dfrac{h}{r}\right) = n_2\left(\dfrac{h}{r} - \dfrac{h}{y}\right)$

4. $\dfrac{n_1}{x} + \dfrac{n_2}{y} = \dfrac{n_2 - n_1}{r}$
   Cancel h. The result is independent of h, so all paraxial rays from O pass through I: I is the image.

5. Cartesian signs: $u = -x$, $v = +y$, $R = +r$

ANSWER $\dfrac{n_2}{v} - \dfrac{n_1}{u} = \dfrac{n_2 - n_1}{R}$

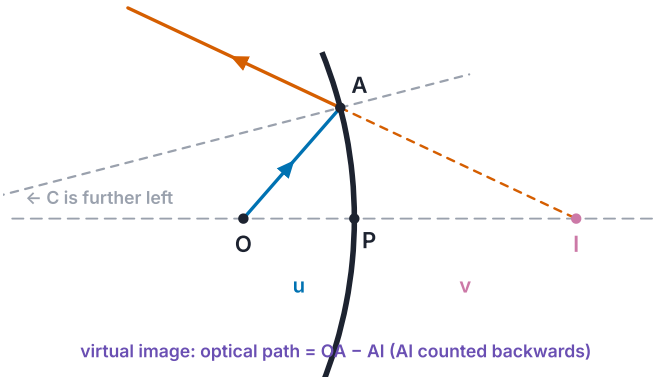
Marks: figure 1 · geometry and paraxial step 2 · Snell + result 1 · signs and conclusion 1.

26 PYQ N20 Q25

Mirror equation by Fermat, concave mirror, object closer than $R/2$.

KNOW FIRST

T10 Reflection at a spherical mirror T8 The same formula from Fermat (optical path route) T2 Fermat's principle: stationary, not "least" M1 Radians and the small-angle trick M4 The binomial shortcut



Fermat for the concave mirror with object inside $R/2$ (PYQ N20 Q25). The image is virtual, so the "path" to I is measured backwards along the dashed line.

Let the mirror have centre C and radius r. O is at distance $x < r/2$ from the pole P, so the image Q is virtual, behind the mirror, at distance y. S is a point on the mirror with $\angle PCS = \theta$.

1. $OS^2 = r^2 + (r-x)^2 - 2r(r-x)\cos\theta$
   Cosine rule in triangle OCS, with $CO = r - x$.

2. $OS \approx x + \frac{1}{2}r^2\left(\frac{1}{x} - \frac{1}{r}\right)\theta^2$
   $\cos\theta \approx 1 - \theta^2/2$, then $\sqrt{1+\varepsilon} \approx 1 + \varepsilon/2$ (paraxial).

3. $SQ^2 = r^2 + (r+y)^2 - 2r(r+y)\cos\theta$, so $SQ \approx y + \frac{1}{2}r^2\left(\frac{1}{y} + \frac{1}{r}\right)\theta^2$
   Triangle QCS, with $CQ = r + y$.

4. The reflected ray only appears to come from Q, so the relevant optical path is $L = OS - SQ$
   As for any virtual image: for a far point B on the reflected ray, $OS + SB = (OS - SQ) + (QS + SB)$, and $QS + SB$ is stationary when Q, S, B are collinear.

5. $L \approx (x-y) + \frac{1}{2}r^2\left[\frac{1}{x} - \frac{1}{y} - \frac{2}{r}\right]\theta^2$

6. Fermat, $\frac{dL}{d\theta} = 0$ for all paraxial θ: $\frac{1}{x} - \frac{1}{y} = \frac{2}{r}$
   Coefficient of θ must vanish.

7. Signs: $u = -x$, $v = +y$ (behind the mirror), $R = -r$ (C to the left)

ANSWER $\frac{1}{u} + \frac{1}{v} = \frac{2}{R}$

Also, $\frac{1}{y} = \frac{1}{x} - \frac{2}{r} > 0$ requires $x < \frac{r}{2}$, confirming the image is virtual for an object inside the focus.

Marks: figure 1 · OS, SQ approximations 2 · Fermat condition 1 · signs/result 1.

27 PYQ N25 Q13

Biconvex, $n = 1.5$, radii 50 cm and 100 cm. Image at 60 cm. Find the object distance.

KNOW FIRST

T11 Thin lens formula and lens-maker's formula P4 The sign rule (Cartesian convention) P3 Mirrors and lenses from Plus Two M6 Working with reciprocals

1. $R_1 = +50$ cm, $R_2 = -100$ cm
   Biconvex lens, light from the left.

2. $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 0.5\left(\frac{1}{50} + \frac{1}{100}\right) = 0.015\ \text{cm}^{-1}$
   Lens-maker's formula.

3. $f = 66.7$ cm
   Converging lens.

4. The image distance (60 cm) is less than f. A converging lens forms a real image of a real object only beyond f, so the image must be virtual, on the object's side: $v = -60$ cm

5. $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{60} - \frac{1}{66.7} = -0.01667 - 0.015 = -0.03167$
   Thin lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$.

6. $u = -31.6$ cm
   The object is 31.6 cm in front of the lens, inside F_1 (magnifier). Magnification $m = v/u = 1.9$, upright.

ANSWER **Object 31.6 cm in front of the lens ($u = -31.6$ cm)**

Note for the examiner-minded: if a real image at +60 cm were insisted on, $\frac{1}{u} = \frac{1}{60} - 0.015$ gives $u = +600$ cm, a virtual object (a converging beam). Mention it in one line; the expected answer is 31.6 cm.

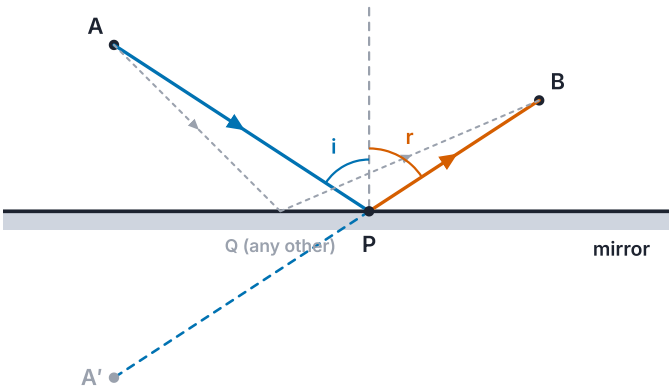
Marks: signs of R 1 · f 1 · reasoning for v 1 · calculation 1 · answer with nature 1.

28

Laws of reflection from Fermat's principle.

KNOW FIRST

T3 Result 1 and 2: straight lines and the law of reflection T2 Fermat's principle: stationary, not "least" M2 Geometry tools



Fermat → law of reflection. A′ is the mirror image of A, so AP = A′P. The path A′PB is shortest when it is a **straight line**, which forces $i = r$. Any other point Q gives a longer path.

1. Light goes from A to B after reflection at a point P of the plane mirror. Let A′ be the image of A in the mirror (AA′ ⊥ mirror, bisected by it)
   Construction.

2. For every point P on the mirror, $AP = A'P$, so $AP + PB = A'P + PB$
   The mirror is the perpendicular bisector of AA′.

3. By Fermat's principle the actual path is stationary (here, least). $A'P + PB$ is least when A′, P, B are collinear
   Any other point Q gives $A'Q + QB > A'B$.

4. The line A′B lies in the plane containing A, A′, B. Since AA′ is parallel to the normal at P, the normal also lies in this plane
   So the incident ray, reflected ray and normal are coplanar (first law).

5 Since A′PB is straight, the angle of PB with the normal equals the angle of PA′ with the normal produced (vertically opposite), which by symmetry equals the angle of PA with the normal

So $i = r$ (second law).

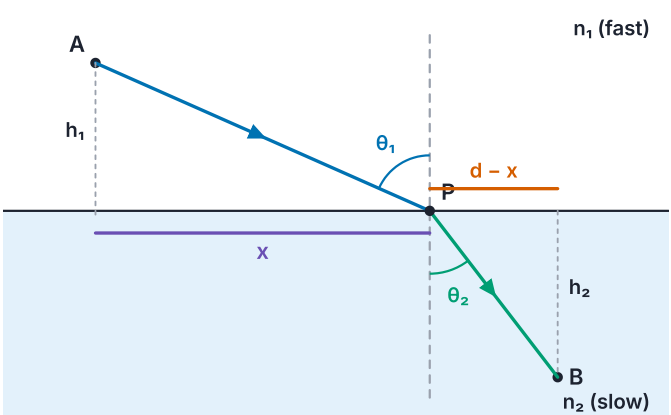
ANSWER **Coplanar, and $i = r$. The path is a minimum.**

29

Snell's law from Fermat's principle.

KNOW FIRST

T4 Result 3: Snell's law from Fermat T2 Fermat's principle: stationary, not "least" M3 Slope zero: finding a stationary point M2 Geometry tools



Fermat → Snell. Slide P along the surface (change x). The optical path $n_1 \cdot AP + n_2 \cdot PB$ is stationary when $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

A (in medium n_1) is at height h_1 and B (in n_2) at depth h_2 from the plane surface. Their feet are a

distance d apart. The ray crosses the surface at P, a distance x from the foot of A.

1. $L(x) = n_1\sqrt{x^2 + h_1^2} + n_2\sqrt{(d-x)^2 + h_2^2}$
   Optical path.

2. $\dfrac{dL}{dx} = \dfrac{n_1 x}{\sqrt{x^2 + h_1^2}} - \dfrac{n_2(d-x)}{\sqrt{(d-x)^2 + h_2^2}} = 0$
   Fermat: optical path stationary.

3. $\sin\theta_1 = \dfrac{x}{\sqrt{x^2 + h_1^2}}, \quad \sin\theta_2 = \dfrac{d-x}{\sqrt{(d-x)^2 + h_2^2}}$
   From the figure, angles from the normal.

4. $n_1 \sin\theta_1 = n_2 \sin\theta_2$

If P is moved out of the plane containing A, B and the normal, both segments lengthen, so the actual path lies in that plane: incident ray, refracted ray and normal are coplanar. Also $d^2L/dx^2 > 0$, so the time is a minimum.

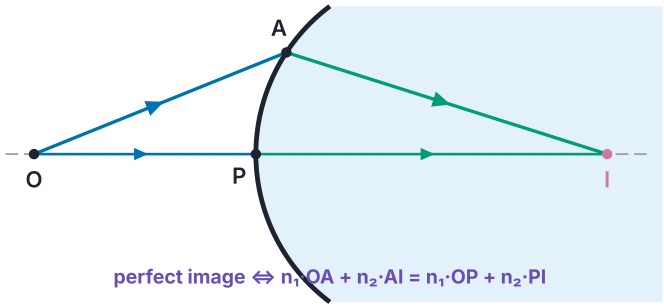
ANSWER $n_1 \sin\theta_1 = n_2 \sin\theta_2$**, rays and normal coplanar**

30

Equal optical paths ⇒ perfect image; spherical surface perfect only for paraxial rays.

KNOW FIRST

T5 Special mirrors and perfect images **T8** The same formula from Fermat (optical path route) **T2** Fermat's principle: stationary, not "least"



Equal optical paths. If every ray from O reaches I with the same optical path, all of them arrive together and I is a sharp image.

A point I is a perfect image of O if every ray leaving O passes through I. By Fermat's principle, each such ray must have a stationary optical path. If many neighbouring paths all reach I, the optical path must be the **same** for all of them:

$$n_1\,OA + n_2\,AI = n_1\,OP + n_2\,PI$$

for every point A of the surface.

Then all the light from O arrives at I in step (same time), giving a sharp image. Surfaces that satisfy this exactly are Cartesian ovals: a paraboloid for parallel light, an ellipsoid for its two foci.

For a spherical surface (T8) the optical path from O through a point at angle θ is

$$L \approx (n_1 x + n_2 y) + \tfrac{1}{2} r^2 \left[\cdots\right] \theta^2 + (\theta^4 \text{ terms})$$

where the bracket is $\left[\dfrac{n_1}{x} + \dfrac{n_2}{y} - \dfrac{n_2 - n_1}{r}\right]$.

At the image the θ^2 term vanishes, but the θ^4 terms do not. So the optical paths are equal only when θ is small, i.e. for paraxial rays. Rays far

from the axis do not meet at I: spherical aberration.

31

Condition for a virtual image by Fermat's principle.

KNOW FIRST

T8 The same formula from Fermat (optical path route) T2 Fermat's principle: stationary, not "least" P3 Mirrors and lenses from Plus Two

Consider refraction at a surface (from n_1 to n_2) where the refracted rays diverge. Let I be a point on the object's side, S any point on the surface, and B any point on a refracted ray, far away.

1. Actual optical path: $L = n_1\,OS + n_2\,SB$
2. Add and subtract $n_2\,SI$: $L = (n_1\,OS - n_2\,SI) + n_2(IS + SB)$
3. Choose I such that $n_1\,OS - n_2\,SI$ is independent of S (Ghatak 3.11)
   The condition for a virtual image at I.
4. Then Fermat requires only $IS + SB$ to be stationary. That happens when I, S and B are collinear
   A straight line is the shortest path from I to B.
5. So every refracted ray, produced backwards, passes through I: the rays appear to diverge from I
   This is the meaning of a virtual image (Ghatak 3.12).

ANSWER $n_1 OS - n_2 SI =$ **constant for all S; the refracted rays appear to come from I**

The minus sign means the path from S to I is counted backwards, along a line light does not actually travel.

32

Air → glass ($n = 1.5$): three cases.

KNOW FIRST

T7 Refraction at one spherical surface (geometry route) T9 Foci of one surface, and surfaces in a row P4 The sign rule (Cartesian convention) M6 Working with reciprocals

Use $\frac{n_2}{v} = \frac{n_2 - n_1}{R} + \frac{n_1}{u}$ with $n_1 = 1$, $n_2 = 1.5$.

1. **(a)** $R = +10$, $u = -30$: $\frac{1.5}{v} = \frac{0.5}{10} - \frac{1}{30} = 0.0167$, so $v = +90$ cm
   Real image in the glass, 90 cm from the pole.

2. **(b)** $R = +10$, converging beam towards a point 20 cm behind: virtual object $u = +20$: $\frac{1.5}{v} = 0.05 - 0.05 = 0$, so $v = \infty$
   The refracted beam is parallel to the axis.

3. **(c)** $R = -10$, $u = -30$: $\frac{1.5}{v} = \frac{0.5}{-10} - \frac{1}{30} = -0.0833$, so $v = -18$ cm
   Virtual image, 18 cm in front of the surface (on the air side).

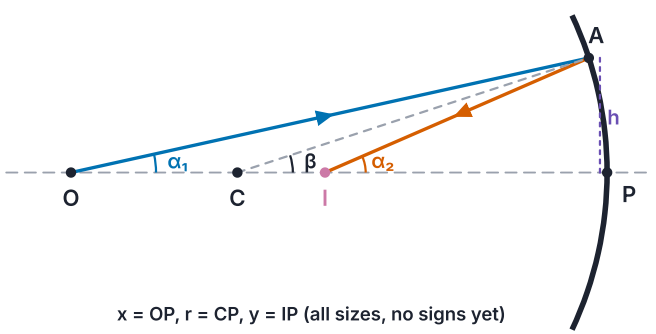
ANSWER (a) +90 cm, real · (b) at infinity (parallel beam) · (c) −18 cm, virtual

33

Mirror equation from the law of reflection (paraxial).

KNOW FIRST

T10 Reflection at a spherical mirror T6 Paraxial rays, axis and signs M2 Geometry tools M1 Radians and the small-angle trick



Concave mirror (Ghatak Fig. 4.3 style). Grey dashed CA is the normal at A. The incident ray OA and reflected ray AI make equal angles with it.

Concave mirror, centre C, radius r. Object O at distance x, image I at distance y from the pole P. A ray OA strikes at A (height h). CA is the normal. $\phi_1 = \angle OAC$, $\phi_2 = \angle CAI$.

1. Triangle OAC: $\beta = \alpha_1 + \phi_1$, so $\phi_1 = \beta - \alpha_1$
   Exterior angle at C.
2. Triangle CAI: $\alpha_2 = \beta + \phi_2$, so $\phi_2 = \alpha_2 - \beta$
   Exterior angle at I.

3. Law of reflection $\phi_1 = \phi_2$: $2\beta = \alpha_1 + \alpha_2$

4. Paraxial: $\beta \approx \frac{h}{r}$, $\alpha_1 \approx \frac{h}{x}$, $\alpha_2 \approx \frac{h}{y}$, so $\frac{2}{r} = \frac{1}{x} + \frac{1}{y}$

5. Signs: O, I, C to the left of P, so $u = -x$, $v = -y$, $R = -r$

ANSWER $\frac{1}{u} + \frac{1}{v} = \frac{2}{R}$, with $f = R/2$

34

(a) Apparent depth; (b) glass hemisphere.

KNOW FIRST

T7 Refraction at one spherical surface (geometry route) T9 Foci of one surface, and surfaces in a row P4 The sign rule (Cartesian convention)

(a) For a plane surface $R \to \infty$, so $\frac{n_2}{v} - \frac{n_1}{u} = 0$, i.e. $v = \frac{n_2}{n_1}u$. Light goes from the liquid ($n_1 = n$) into air ($n_2 = 1$). The coin is at $u = -d$:

$$v = \frac{1}{n}(-d) = -\frac{d}{n}$$

So the coin appears at depth d/n below the surface (a virtual image, raised up).

(b) Light falls on the flat face first.

1. Flat face: $R = \infty$, object at infinity, so the image is at infinity
   The beam stays parallel inside the glass.

2. Curved face, bulging towards the outgoing light, so its centre is to the left: $R = -20$ cm. From glass ($n_1 = 1.5$) to air ($n_2 = 1$), $u = -\infty$
   Sign rule.

3. $\frac{1}{v} = \frac{1 - 1.5}{-20} = 0.025$, so $v = +40$ cm

ANSWER **(a) apparent depth d/n · (b) focus 40 cm beyond the curved face**

35

Concave mirror, $R = 60$ cm.

KNOW FIRST
T10 Reflection at a spherical mirror T15 Lateral magnification P4 The sign rule (Cartesian convention) M6 Working with reciprocals

Concave mirror facing the light: $R = -60$ cm, $f = -30$ cm. Use $\frac{1}{v} = \frac{2}{R} - \frac{1}{u}$ and $m = -\frac{v}{u}$.

1. **(a)** $u = -80$: $\frac{1}{v} = -\frac{1}{30} + \frac{1}{80} = \frac{-8 + 3}{240}$, so $v = -48$ cm
   In front of the mirror: real.

2. $m = -\frac{-48}{-80} = -0.6$
   Inverted, 0.6 times the size.

3. **(b)** The image falls on the object when $u = v$: $\frac{2}{u} = \frac{2}{R}$, so $u = R = -60$ cm
   At the centre of curvature: rays along the normal retrace themselves. $m = -1$.

4 **(c)** $u = -20$: $\frac{1}{v} = -\frac{1}{30} + \frac{1}{20} = \frac{1}{60}$, so $v = +60$ cm, $m = -\frac{60}{-20} = +3$

Behind the mirror: virtual, upright, 3 times bigger (object inside F).

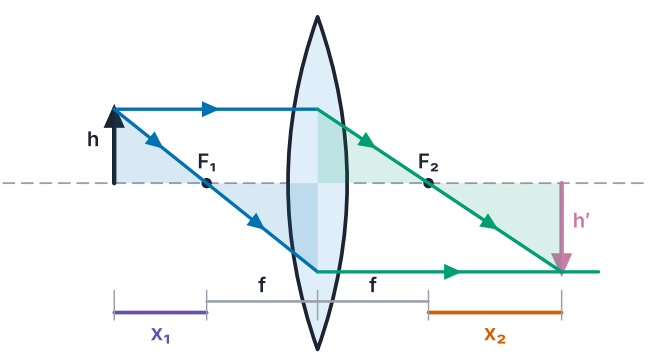
ANSWER **(a) 48 cm in front, real, inverted, m = −0.6 · (b) at C, 60 cm · (c) 60 cm behind, virtual, erect, m = +3**

36

Derive Newton's formula for a thin lens.

KNOW FIRST

T14 Newton's formula T13 Finding the image by drawing M2 Geometry tools P4 The sign rule (Cartesian convention)



Newton's formula. x_1 is measured from F_1 to the object, x_2 from F_2 to the image (not from the lens!). Blue triangles are similar; green triangles are similar.

A thin convex lens of focal length f is in air. An object of height h is at distance x_1 from F_1; its

image, of height h', is at distance x_2 from F_2.

1. Ray through F_1 emerges parallel at the height of the image top. Similar triangles at F_1: $\dfrac{|h|}{|x_1|} = \dfrac{|h'|}{f}$
2. Ray parallel to the axis passes through F_2. Similar triangles at F_2: $\dfrac{|h|}{f} = \dfrac{|h'|}{|x_2|}$
3. Equating the two values of $|h'|/|h|$: $\dfrac{f}{|x_1|} = \dfrac{|x_2|}{f}$, so $|x_1||x_2| = f^2$
4. With signs, the object is left of F_1 ($x_1 < 0$) and the image right of F_2 ($x_2 > 0$): $x_1 x_2 = -f^2$. In general, $x_1 x_2 = f_1 f_2$

ANSWER $x_1 x_2 = f_1 f_2 = -f^2$ **(magnitudes:** $|x_1||x_2| = f^2$**)**

Also, from steps 1–2: $m = \dfrac{h'}{h} = \dfrac{f}{x_1} = -\dfrac{x_2}{f}$.

37

Biconvex: $R_1 = 100$, $R_2 = 60$, $n = 1.5$, object 1 cm tall at 100 cm.

KNOW FIRST

T11 Thin lens formula and lens-maker's formula T14 Newton's formula T15 Lateral magnification P4 The sign rule (Cartesian convention) M6 Working with reciprocals

1. $R_1 = +100$, $R_2 = -60$: $\frac{1}{f} = 0.5\left(\frac{1}{100} + \frac{1}{60}\right) = 0.5 \times \frac{8}{300} = \frac{1}{75}$, so $f = 75$ cm

2. $u = -100$: $\frac{1}{v} = \frac{1}{75} - \frac{1}{100} = \frac{4-3}{300}$, so $v = +300$ cm
   Real image, 300 cm behind the lens.

3. $m = \frac{v}{u} = \frac{300}{-100} = -3$, so $h' = -3$ cm
   Inverted, 3 cm tall.

4. $x_1 = u - f_1 = -100 - (-75) = -25$ cm, $x_2 = v - f_2 = 300 - 75 = +225$ cm
   Distances from the foci ($f_1 = -75$, $f_2 = +75$).

5. $x_1 x_2 = (-25)(225) = -5625 = -(75)^2 = -f^2$ ✓
   Newton's formula verified. Also $m = f/x_1 = 75/(-25) = -3$ ✓.

ANSWER **f = 75 cm · v = +300 cm · image 3 cm, inverted · $x_1x_2 = -f^2$**

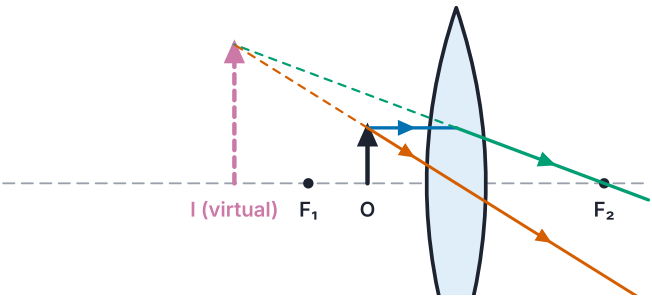
38

Object 12 cm from a convex lens of $f = 20$ cm.

KNOW FIRST

T13 Finding the image by drawing T11 Thin lens formula and lens-maker's formula T15 Lateral magnification P3 Mirrors and lenses from Plus Two

object inside F_1 → rays spread out after the lens;
traced back (dashed) they meet at I: upright, 2.5× bigger



Drawn to scale. Ray 1 (parallel → through F_2) and ray 2 (through the centre) diverge. Their backward extensions meet at I.

1. Construction: ray 1 (parallel → through F_2) and ray 2 (through the centre) diverge after the lens. Produced backwards they meet on the object's side

   Virtual image, upright, enlarged.

2. $\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{20} - \frac{1}{12} = \frac{3-5}{60} = -\frac{1}{30}$, so $v = -30$ cm

3. $m = \frac{v}{u} = \frac{-30}{-12} = +2.5$

4. Check with Newton: $x_1 = -12 + 20 = +8$, $x_2 = -30 - 20 = -50$, $x_1 x_2 = -400 = -f^2$ ✓

ANSWER **Image 30 cm in front of the lens (same side), virtual, erect, 2.5 times enlarged**

Section C · Essays
10 marks each. About two pages: statement/theory (3–4), derivation with figure (3–4), discussion/result/example (2–3).

39

Fermat's principle → refraction at a spherical surface (optical path method); maximum or minimum?

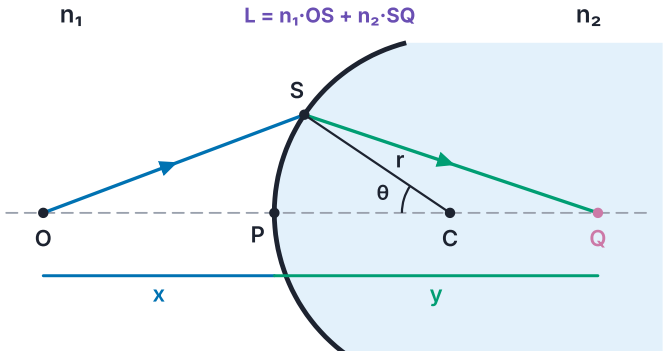
KNOW FIRST

T2 Fermat's principle: stationary, not "least" T8 The same formula from Fermat (optical path route) T7 Refraction at one spherical surface (geometry route) M1 Radians and the small-angle trick M3 Slope zero: finding a stationary point M4 The binomial shortcut P4 The sign rule (Cartesian convention)

1. Fermat's principle

The optical path of a ray from A to B is $L = \int_A^B n\,ds = c\tau$, where τ is the time taken. Fermat's principle states that the actual ray is the path for which the optical path is **stationary** with respect to small variations of the path, $\delta \int n\,ds = 0$. It may be a minimum, a maximum or constant.

2. Set-up



Fermat route (Ghatak Fig. 3.10). Move S around the surface by changing θ. The ray is real when L is stationary: dL/dθ = 0.

A spherical surface, centre C, radius r, separates media n_1 and n_2. O is a point object at distance x from the pole P. S is a point on the surface with $\angle SCP = \theta$. Q is a point on the axis in medium 2, at distance y from P.

3. Optical path O → S → Q

1. $OS^2 = (x+r)^2 + r^2 - 2r(x+r)\cos\theta$
   Cosine rule in triangle OCS.

2. With $\cos\theta \approx 1 - \dfrac{\theta^2}{2}$ (paraxial): $OS^2 \approx x^2 + r(x+r)\theta^2$

3. $OS \approx x + \frac{1}{2}r^2\left(\dfrac{1}{r} + \dfrac{1}{x}\right)\theta^2$
   Binomial approximation.

4. Similarly, in triangle SCQ ($CQ = y - r$, angle at C $= \pi - \theta$): $SQ \approx y - \frac{1}{2}r^2\left(\dfrac{1}{r} - \dfrac{1}{y}\right)\theta^2$

5. $L = n_1 OS + n_2 SQ \approx (n_1 x + n_2 y) + \frac{1}{2} r^2 \left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r} \right] \theta^2$

4. Applying Fermat's principle

1. $\frac{dL}{d\theta} = r^2 \left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r} \right] \theta = 0$

2. $\theta = 0$ gives only the axial ray OPQ, which passes undeviated (normal incidence). For **all** paraxial rays from O to reach the same point, the bracket must vanish

3. $\frac{n_1}{x} + \frac{n_2}{y_0} = \frac{n_2 - n_1}{r}$
   This fixes the image distance y_0. All paraxial rays take equal time from O to I.

4. Signs $u = -x$, $v = y_0$, $R = r$: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

5. Maximum or minimum?

1. $\frac{d^2 L}{d\theta^2} = r^2 \left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r} \right]$

2. Using the image condition, $\frac{n_2 - n_1}{r} - \frac{n_1}{x} = \frac{n_2}{y_0}$: $\frac{d^2 L}{d\theta^2} = r^2 n_2 \left(\frac{1}{y} - \frac{1}{y_0} \right)$

3. Q beyond the image ($y > y_0$): $\frac{d^2 L}{d\theta^2} < 0$, so the actual ray OPQ is a path of **maximum** optical path

4. Q between P and I ($y < y_0$): $\dfrac{d^2L}{d\theta^2} > 0$, a **minimum**

5. Q at I ($y = y_0$): L is independent of θ (to this order), so it is stationary, neither a maximum nor a minimum

Physical check: $n_1OP + n_2PI = n_1OS + n_2SI$. For Q beyond I, $n_1OP + n_2PQ = n_1OS + n_2(SI + IQ) > n_1OS + n_2SQ$, because $SI + IQ > SQ$ (triangle inequality). The axial path is longer than its neighbours: a maximum.

6. Conclusion

Fermat's principle gives the refraction formula without using Snell's law. It also shows that the real ray can be a path of maximum time, so the principle must be stated as "stationary", not "least" time.

40

Thin lens formula and lens-maker's formula; assumptions; converging vs diverging.

KNOW FIRST

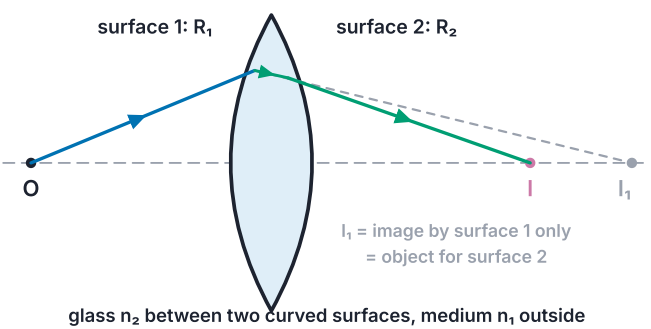
T11 Thin lens formula and lens-maker's formula T7 Refraction at one spherical surface (geometry route) T9 Foci of one surface, and surfaces in a row T12 Principal foci and focal length P4 The sign rule (Cartesian convention)

1. Refraction at one surface

For a spherical surface separating n_1 (incident side) and n_2: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ (Cartesian signs, paraxial rays).

2. Assumptions

- The lens is thin: its thickness is negligible, so all distances are measured from the centre P.
- Rays are paraxial.
- The lens (index n_2) has the same medium (index n_1) on both sides.



A thin lens = two curved surfaces very close together. Surface 1 alone would form an image I_1. Surface 2 takes I_1 as its object and forms the final image I.

3. Derivation

1. First surface (radius R_1), from n_1 into n_2:
   $\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$
   If the second surface were absent, the image would be at v'.

2. This image acts as the object for the second surface (radius R_2), from n_2 into n_1: $\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2}$
   Thin lens, so v' is measured from the same point.

3. Adding: $\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

4. Dividing by n_1, with $n = n_2/n_1$: $\frac{1}{v} - \frac{1}{u} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

5. For $u \to -\infty$ the image is at F_2, $v = f$: $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
   Lens-maker's formula.

ANSWER $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

4. Converging or diverging

- $f > 0$: converging; $f < 0$: diverging.
- Lens denser than its surroundings ($n > 1$): biconvex ($R_1 > 0$, $R_2 < 0$) gives $f > 0$, converging; biconcave gives $f < 0$, diverging.
- Lens rarer than its surroundings ($n < 1$): the roles reverse. A biconvex lens diverges and a biconcave lens converges (e.g. an air bubble in water).

Example: $n = 1.5$, $R_1 = 20$ cm, $R_2 = -20$ cm in air: $\frac{1}{f} = 0.5 \times \frac{2}{20}$, so $f = 20$ cm. In water ($n = 1.128$) $f \approx 78$ cm, still converging.

41

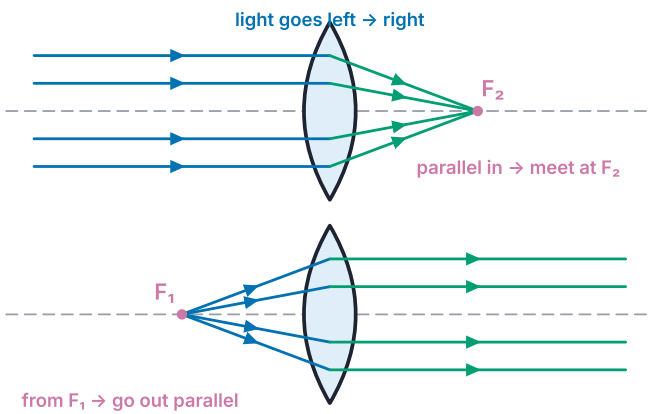
Principal foci; f_1, f_2 for different media; $f_2 = -f_1$; lens in a denser medium.

KNOW FIRST

T12 Principal foci and focal length T11 Thin lens formula and lens-maker's formula T9 Foci of one surface, and surfaces in a row T7 Refraction at one spherical surface (geometry route)

1. Definitions

First principal focus F_1: the axial point such that rays from it (or directed towards it) emerge parallel to the axis; its image is at infinity. **Second principal focus F_2:** the axial point at which rays incident parallel to the axis meet (or from which they appear to diverge) after refraction. f_1 and f_2 are their distances from the lens, with sign.



Top: **second principal focus F_2**: rays parallel to the axis meet there after the lens. Bottom: **first principal focus F_1**: rays starting there come out parallel. For a thin lens in air both are a distance f from the lens.

2. Lens between different media

Medium n_1 on the left, lens of index n_2, medium n_3 on the right.

1. Surface 1: $\dfrac{n_2}{v'} - \dfrac{n_1}{u} = \dfrac{n_2 - n_1}{R_1}$
2. Surface 2 (from n_2 to n_3): $\dfrac{n_3}{v} - \dfrac{n_2}{v'} = \dfrac{n_3 - n_2}{R_2}$
3. Add: $\dfrac{n_3}{v} - \dfrac{n_1}{u} = \dfrac{n_2 - n_1}{R_1} + \dfrac{n_3 - n_2}{R_2} \equiv D$
4. F_1: $v = \infty$, $u = f_1$: $\dfrac{1}{f_1} = -\dfrac{1}{n_1}\left[\dfrac{n_2 - n_1}{R_1} + \dfrac{n_3 - n_2}{R_2}\right]$
5. F_2: $u = -\infty$, $v = f_2$: $\dfrac{1}{f_2} = \dfrac{1}{n_3}\left[\dfrac{n_2 - n_1}{R_1} + \dfrac{n_3 - n_2}{R_2}\right]$
6. Hence $\dfrac{f_2}{f_1} = -\dfrac{n_3}{n_1}$

3. Same medium on both sides

With $n_3 = n_1$: $\dfrac{f_2}{f_1} = -1$, so $f_2 = -f_1 = f$, where $\dfrac{1}{f} = \left(\dfrac{n_2}{n_1} - 1\right)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$. The two foci are equidistant from the lens, on opposite sides. For a converging lens $f_1 < 0$, $f_2 > 0$; for a diverging lens $f_1 > 0$, $f_2 < 0$.

4. Biconvex lens in a denser medium

If $n_1 > n_2$ (e.g. glass of 1.5 in a liquid of 1.6), then $n = n_2/n_1 < 1$, so $n - 1 < 0$. With $\dfrac{1}{R_1} - \dfrac{1}{R_2} > 0$, f

becomes negative: the biconvex lens **diverges**. Example: $n = 1.5/1.6 = 0.9375$, so the lens is a weak diverging lens.

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Lateral magnification: single surface, thin lens, Newtonian form, example.

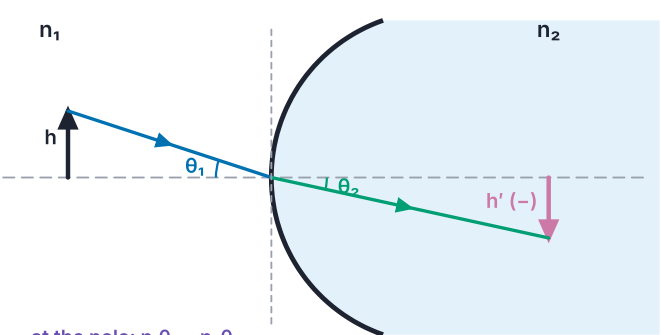
KNOW FIRST

T15 Lateral magnification T14 Newton's formula T7 Refraction at one spherical surface (geometry route) T11 Thin lens formula and lens-maker's formula M1 Radians and the small-angle trick M2 Geometry tools

1. Definition

Lateral magnification $m = h'/h$: the ratio of image height to object height, both measured perpendicular to the axis with sign. $m > 0$ erect, $m < 0$ inverted.

2. Single refracting surface



Magnification at a single surface: follow the ray that hits the pole. $\theta_1 \approx h/(-u)$, $\theta_2 \approx -h'/v$, and Snell (small angle) gives $m = h'/h = n_1 v/(n_2 u)$.

1. Take the ray from the top of the object to the pole P. At P the normal to the surface is the axis itself. Angles of incidence and refraction: $\theta_1 \approx \dfrac{h}{-u}$, $\theta_2 \approx \dfrac{-h'}{v}$
   Paraxial; $-u$ and $-h'$ are positive.
2. Snell (small angles): $n_1\theta_1 = n_2\theta_2$, so $n_1\dfrac{h}{-u} = n_2\dfrac{-h'}{v}$
3. $m = \dfrac{h'}{h} = \dfrac{n_1 v}{n_2 u}$

3. Thin lens as two surfaces

1. First surface: $m_1 = \dfrac{n_1 v'}{n_2 u}$; second surface: $m_2 = \dfrac{n_2 v}{n_1 v'}$
2. Total: $m = m_1 m_2 = \dfrac{v}{u}$

4. In terms of distances from the foci

From the similar triangles of Newton's formula (with x_1 from F_1, x_2 from F_2):

$$m = -\frac{f_1}{x_1} = -\frac{x_2}{f_2} \quad \Rightarrow \quad m = \frac{f}{x_1} = -\frac{x_2}{f} \text{ (lens in air)}$$

5. Numerical example

Surface: air to glass ($n = 1.5$), $R = +10$ cm, $u = -30$ cm gives $v = +90$ cm, so $m = \dfrac{1 \times 90}{1.5 \times (-30)} = -2$: inverted, twice as big.

Lens: $f = 75$ cm, $u = -100$ cm gives $v = +300$ cm, so $m = v/u = -3$. With $x_1 = -25$ cm: $m = f/x_1 =$

-3 ✓; with $x_2 = 225$ cm: $m = -x_2/f = -3$ ✓.

ANSWER

Surface	$m = \dfrac{n_1 v}{n_2 u}$
Thin lens	$m = \dfrac{v}{u} = \dfrac{f}{x_1} = -\dfrac{x_2}{f}$
Mirror	$m = -\dfrac{v}{u}$